

Charge, covariance, and ensemble convergence in the class-A adaptive circuit

Internal campaign summary¹

¹ *Class-A fermionic adaptive-circuit project*
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This note consolidates the charge-fluctuation and covariance-convergence analysis of Born-rule class-A adaptive-circuit trajectories with a dynamical domain wall. The principal production ensembles have $N_x = 20$, $N_y = 30, 40, 50$, two orbitals per unit cell, 100 trajectories, $n_{\text{shell}} = 1$, hard domain-wall truncation, and $2N_y$ saved cycles. We distinguish three logically different questions: purification of a maximally mixed initial state, motion and stationarity of pure trajectories, and statistical independence of cycle-resolved samples. Global charge variance and the root-mean-square normalized covariance norm reach an operational transient threshold after only 15–16 cycles, with no detectable growth over the available N_y range. Spatial entropy contours, however, reveal a fast active-bulk contraction with intrinsic time 0.56–0.58 cycles and an algebraic wall tail $s_w \sim t^{-\alpha_w}$ with $\alpha_w = 1.12$ –1.15. The wall reaches 0.01 bits per cell only after 46–57 cycles. Late-time trajectory-to-trajectory occupancy variance is 97.2–97.3% concentrated on the exact wall rows, whereas the within-trajectory quantum variance is not wall dominated. Finally, block-stationarity and autocorrelation tests on the default pure-state ensemble show that post-burn-in cycles are best viewed as samples from a *stationary, temporally correlated* ensemble. The charge-density autocorrelation time is only 1.3–1.4 cycles, while the long-range correlator exponent has $\tau_{\text{int}} = 14.8, 20.3, 24.1$ cycles, approximately $0.49N_y$. Thus every retained cycle may contribute to an ensemble estimate once stationarity is demonstrated, but neighboring cycles must not be counted as independent samples.

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I. SCOPE AND BOTTOM LINE

The original sequence of questions concerned charge variance, its cycle derivative, Born-rule total-charge traces, Frobenius norms of covariance matrices, spatial wall-versus-bulk resolution, and a defensible burn-in rule.

Those questions involve several different random variables and two different initial-state ensembles. The first purpose of this note is to prevent them from being conflated.

There is a domain wall in every principal run analyzed here. The $N_x = 20$ production geometry has walls at $x = 5$ and $x = 15$, a topological active slab $5 \leq x \leq 15$, and hard truncation of overcomplete Wannier (OW) modes across the interfaces. The run parameters are $(\alpha_1, \alpha_2) = (1, 30)$, $n_{\text{shell}} = 1$, ancillary filling $\bar{n}_a = 1/2$, and the `perfect_correction` protocol with `postselect=False`. A full cycle consists of OW-number measurements with conditional fermionic-swap feedforward, followed by incoherent particle redistribution and ancillary occupation measurement, as in Ref. [1]. Distinct OW stabilizers need not commute, and their localized modes form an overcomplete rather than orthonormal Wannier basis.

The conclusions are observable dependent:

1. The *global purification transient* is short. A sustained covariance-norm deficit below 10^{-3} together with a normalized norm rate below 10^{-4} per cycle occurs at cycles 15, 15, 16 for $N_y = 30, 40, 50$.
2. The *bulk purification transient* is shorter still. The active bulk falls below 0.01 bits per cell at cycle 4 in all three spatial-entropy runs.
3. The *wall purification tail* is parametrically slower and is algebraic over cycles 5–30. Its 0.01-bit threshold occurs at cycles 49, 46, 57 for $16 \times 30, 16 \times 32$, and 30×40 .
4. The rule “burn in for N_y cycles and measure on $[N_y, 2N_y]$ ” is a useful production convention, but it is not a theorem that every observable is stationary at cycle N_y . The entanglement slope is stationary by a paired block test for all three $N_x = 20$ sizes; the long-range correlator exponent still drifts for $N_y = 30, 40$ and is stationary for $N_y = 50$.
5. Stationarity is not independence. Long-wavelength wall observables remain correlated for a substantial fraction of the measurement window. Uncertainty estimates must therefore resample whole trajectories or use autocorrelation-aware blocking rather than treating all trajectory-cycle pairs as independent.

Accordingly, an endpoint measurement at cycle $2N_y$ is well inside the observed low-entropy regime. Pooling every cycle after burn-in is also legitimate for a demonstrated stationary marginal distribution, but only as correlated Monte Carlo sampling. The maximally mixed purification campaign is a special warning: its charge variance continues to decrease through $[N_y, 2N_y]$, so that particular time series is a transient diagnostic rather than a stationary-ensemble sample.

II. GEOMETRY, TRAJECTORY ENSEMBLES, AND PROVENANCE

A. Lattice and canonical correlation convention

Let $\mathbf{r} = (x, y)$ label a unit cell and $\mu \in \{1, 2\}$ its two physical orbitals. The physical fermions are $\hat{c}_{\mathbf{r}, \mu}$, the number of unit cells is $N_{\text{uc}} = N_x N_y$, and the single-particle dimension is

$$V = 2N_{\text{uc}} = 2N_x N_y. \quad (1)$$

The total physical charge is

$$\hat{N}_F = \sum_{\mathbf{r}, \mu} \hat{c}_{\mathbf{r}, \mu}^\dagger \hat{c}_{\mathbf{r}, \mu}. \quad (2)$$

Following the project notation policy, a trajectory-resolved one-particle correlation matrix is

$$G_{ij} = \text{Tr}(\hat{\rho} \hat{c}_i^\dagger \hat{c}_j), \quad 0 \leq G \leq \mathbf{1}_V. \quad (3)$$

The archived simulation helpers call the shifted Hermitian covariance “**G**.” To avoid a collision, this note denotes that archived object by

$$Q = 2G - \mathbf{1}_V, \quad G = \frac{Q + \mathbf{1}_V}{2}. \quad (4)$$

Thus every appearance of $G_t G_{t+1} - \mathbf{1}$ in the exploratory discussion should be read as $Q_t Q_{t+1} - \mathbf{1}_V$ in canonical notation.

B. Born-rule trajectories and temporal samples

For a measurement outcome m with Kraus operator $\hat{K}_m$ and nonnegative weight w_m , the simulation samples

$$p_m = w_m \langle \langle \psi | \hat{K}_m^\dagger \hat{K}_m | \psi \rangle \rangle, \quad \sum_m w_m \hat{K}_m^\dagger \hat{K}_m = \hat{\mathbf{1}}. \quad (5)$$

A full record $\mathbf{m} = (m_1, m_2, \dots)$ defines one member of the many-body evolution-operator (mEO) ensemble $\mathcal{M}_{\text{mEO}} = \{\hat{V}_{\mathbf{m}}\}_{\mathbf{m}}$. The 100 trajectory seeds are independent. Successive cycles within a fixed trajectory are not independent because they form a Markov chain on Gaussian states.

For a trajectory observable $O_s(t)$, the cycle-resolved Born mean is

$$\bar{O}(t) = \frac{1}{S} \sum_{s=1}^S O_s(t), \quad S = 100. \quad (6)$$

If the post-burn-in distribution is stationary, then the marginal distribution of $O_s(t)$ is independent of t and every retained cycle is a valid, though correlated, draw from the same ensemble. This distinction between stationarity and independence is the central statistical point of Sec. IX.

C. Campaign inventory

TABLE I. Data products used in this note. All dynamics were previously run through the canonical GPU entry point `classA.U1FGTN.gpu.run_markov_circuit`; the present work is analysis-only. The entropy-contour sample counts are 10, 10, 25 in the order shown.

Ensemble	Geometry and sampling	Saved information and role
Maximally mixed purification	$20 \times (30, 40, 50)$; $S = 100$; $2N_y$ cycles	Total quantum charge variance, total entropy, local expected charge, local quantum variance, and Chern marker. Used for purification and late-time charge localization.
Default pure-state streaming	$20 \times (30, 40, 50)$; $S = 100$; $2N_y$ cycles	Entanglement and correlator fits, real-space Chern marker, local expected charge, and successive covariance distance. Used for stationarity, autocorrelation, and trajectorywise motion.
Pure-state covariance snapshots	16×30 , 16×40 ; $S = 10$; cycles 5, 10, 20, 50	Full sparse covariance snapshots. Used for the spatially reduced covariance pilot.
Maximally mixed entropy contours	16×30 to cycle 50, 16×32 and 30×40 to cycle 100	Whole-system spatial entropy contours. Used for wall-versus-bulk purification fits and threshold times.

All principal $N_x = 20$ runs use `dw_loc = (5, 15)`, `dw_truncation=True`, `sequence=raster_y`, `complex128` arithmetic, and `postselect=False`. The $N_x = 16$ snapshot constructor places the walls at $x = 4, 12$. Covariance histories were intentionally not retained in the main campaigns; the requested global quantities were reconstructed from streamed scalars, while spatial covariance diagnostics use the separate sparse-snapshot pilot.

D. Prior artifacts: this was not the first look

The question whether rolling charge and Frobenius diagnostics had already been plotted has an affirmative answer, but none of the older artifacts alone contains the complete present comparison. The May 6 notebook `load_purification_charge_fluctuation_data_cpu_edited.ipynb` used the same maximally mixed campaign and displayed rolling centered-charge means and rolling percentage variances, with a different normalization. The older 20×20 notebook `n20x20_dwtrunc_convergence_cpu.ipynb` plotted a successive Frobenius-distance diagnostic for one size. Finally, `diagnose_perfect_correction_slope_excess_cpu.ipynb` used `frob_successive_delta` alongside the entropy-slope excess. The present bundle adds the three-size comparison, the exact normalization identities, Born trace trajectories, spatial charge decomposition, explicit wall/bulk purification fits, and the stationarity/autocorrelation test.

III. OBSERVABLES, NORMALIZATION, AND CONVERGENCE ESTIMATORS

A. Within-trajectory quantum charge variance

For a charge-conserving Gaussian state, Wick's theorem gives

$$\text{Var}_s(\hat{N}_F; t) = \text{tr}[G_s(t) - G_s^2(t)]. \quad (7)$$

This is the quantum variance inside trajectory s . It is not the variance of the trajectory means. The plotted cycle rate is

$$\left| \overline{\partial_t \text{Var}(\hat{N}_F)} \right|(t) = \frac{1}{S} \sum_s \left| \text{gradient}_t \text{Var}_s(\hat{N}_F; t) \right|, \quad (8)$$

where the archived analysis uses `numpy.gradient` on the integer cycle grid with second-order edge handling. Both quantities are shown on logarithmic y axes.

B. Expected total charge and trajectory-to-trajectory variance

The expected total charge on trajectory s is

$$n_s(t) = \text{tr} G_s(t) = \langle \hat{N}_F \rangle_s. \quad (9)$$

Half filling is $V/2 = N_x N_y$, whereas the largest possible charge is $V = 2N_x N_y$. The requested normalized displacement is therefore

$$q_s(t) = \frac{n_s(t) - V/2}{V}. \quad (10)$$

The Born mean $\bar{q}(t)$ measures the ensemble's fractional offset from half filling. Two distinct measures of its spread

are useful:

$$v_N(t) = \frac{\text{Var}_s[n_s(t)]}{V}, \quad (11)$$

$$\sigma_{n/V}(t) = \frac{\sqrt{\text{Var}_s[n_s(t)]}}{V}. \quad (12)$$

The first is an extensive-variance density and is what the trace-variance panel plots. The second is the standard deviation of the fractional filling and is the more direct “typical occupancy deviation.” Dividing the variance only by V should not be confused with the variance of n_s/V , which would contain V^2 in the denominator.

C. Covariance Frobenius norm and the correct size collapse

Equations (4) and (7) imply the exact trajectorywise identity

$$\text{Var}_s(\hat{N}_F; t) = \frac{V - \text{tr } Q_s^2(t)}{4} = \frac{V - \|Q_s(t)\|_F^2}{4}, \quad (13)$$

because Q is Hermitian. Hence

$$\|Q_s(t)\|_F = \sqrt{V - 4 \text{Var}_s(\hat{N}_F; t)}. \quad (14)$$

For a pure number-conserving Gaussian state, $Q^2 = \mathbf{1}_V$ and $\|Q\|_F = \sqrt{V}$. This resolves the normalization puzzle:

$$\frac{\|Q\|_F}{V} \longrightarrow V^{-1/2}, \quad (15)$$

$$\frac{\|Q\|_F}{\sqrt{V}} \longrightarrow 1. \quad (16)$$

Dividing a Frobenius norm by $2N_x N_y = V$ therefore necessarily makes its saturated value smaller at larger N_y : a Frobenius norm is root-extensive, not extensive. Equation (16) is the root-mean-square (RMS) spectral normalization that removes this trivial scaling.

Define the normalized deficit and absolute norm rate by

$$d_F(t) = 1 - \frac{\|Q(t)\|_F}{\sqrt{V}}, \quad (17)$$

$$r_F(t) = \frac{1}{\sqrt{V}} |\partial_t \|Q(t)\|_F|. \quad (18)$$

The displayed r_F uses a centered rolling mean with width five; all threshold tests retain the documented raw trajectory aggregation before applying that same deterministic rolling operation.

D. Adjacent-cycle covariance motion

For the pure-state streaming ensemble, Q_t is a Hermitian involution. Consequently,

$$\begin{aligned} Q_{t-1} Q_t - \mathbf{1}_V &= Q_{t-1} (Q_t - Q_{t-1}), \\ \|Q_{t-1} Q_t - \mathbf{1}_V\|_F &= \|Q_t - Q_{t-1}\|_F. \end{aligned} \quad (19)$$

Left multiplication by Q_{t-1} preserves the Frobenius norm. The streamed scalar `frob_successive_delta` is therefore exactly the requested adjacent-product residual, without saving Q_t itself. A nonzero plateau of this quantity means that trajectories continue moving; it does not mean the ensemble distribution is nonstationary.

E. Spatial reduction and the origin of R_x

Let P_x retain both orbitals at a fixed x row for every y . The principal submatrix

$$Q_x(t) = P_x Q(t) P_x \quad (20)$$

is the shifted covariance matrix of the Gaussian reduced state obtained by tracing out all other x rows. It acts on $2N_y$ modes. The spatially resolved temporal rate introduced in the exploratory analysis is

$$R_x(t_a, t_b) = \frac{\|Q_x(t_b) - Q_x(t_a)\|_F}{(t_b - t_a) \sqrt{2N_y}}. \quad (21)$$

Thus R_x is not a new physical operator. It is simply the finite-difference speed of the reduced covariance on row x , with the same RMS spectral normalization used globally. The partial trace generally makes the reduced state mixed even when the full trajectory is pure. Its mixedness diagnostic is

$$M_x(t) = \frac{\|Q_x^2(t) - \mathbf{1}_{2N_y}\|_F}{\sqrt{2N_y}}. \quad (22)$$

Because M_x need not vanish in a stationary reduced state, the local product residual $\|Q_x(t_a) Q_x(t_b) - \mathbf{1}_{2N_y}\|_F$ would mix temporal motion with ordinary reduced-state mixedness and is not used as a convergence criterion.

F. Regional entropy and convergence thresholds

Let $s(\mathbf{r}, t)$ be the trajectory-averaged entanglement contour in bits. The wall entropy density averages s over the two exact wall rows and all y ; the active-bulk density averages rows inside the topological slab at least two rows away from either wall. The hard-truncated exterior is excluded. The wall is compared on cycles 5–30 to

$$s_w(t) = A_w t^{-\alpha_w}, \quad (23)$$

$$s_w(t) = A_w e^{-t/\tau_w}, \quad (24)$$

using ordinary least squares in log space. The bulk is described by

$$s_b(t) = A_b e^{-t/\tau_b} + B_b t^{-\alpha_w}, \quad (25)$$

where α_w is fixed to the wall fit. A threshold time $T_R(\epsilon)$ is the first saved cycle after which the regional entropy density remains below ϵ for every later saved cycle.

G. Stationarity and autocorrelation estimators

For each observable $O_s(t)$, the measurement window is the inclusive set $N_y \leq t \leq 2N_y$. It is divided at $3N_y/2$. The paired block difference for trajectory s is

$$d_s = \langle O_s \rangle_{[3N_y/2, 2N_y]} - \langle O_s \rangle_{[N_y, 3N_y/2]}. \quad (26)$$

The reported 95% interval for $\bar{d}$ is a percentile interval from 2000 resamplings of complete trajectories. This is a direct test for drift between the early and late halves of the proposed measurement window.

To estimate temporal dependence without confusing it with common drift, define residuals

$$\delta O_s(t) = O_s(t) - \bar{O}(t). \quad (27)$$

Their pooled lag correlation is

$$\rho_O(k) = \frac{\sum_{s,t} \delta O_s(t) \delta O_s(t+k)}{\sqrt{\sum_{s,t} \delta O_s^2(t) \sum_{s,t} \delta O_s^2(t+k)}}. \quad (28)$$

The integrated autocorrelation time uses the initial-positive sequence,

$$\tau_{\text{int}}[O] = 1 + 2 \sum_{k=1}^{k_+} \rho_O(k), \quad (29)$$

where k_+ ends immediately before the first nonpositive lag. The approximate number of independent-equivalent samples is

$$N_{\text{eff}}[O] \simeq \frac{S(N_y + 1)}{\tau_{\text{int}}[O]}. \quad (30)$$

This is a diagnostic, not a proof that every functional of the full covariance has equilibrated.

TABLE II. Normalization dictionary. The archive calls the shifted covariance Q by the name $\mathbb{G}$; Eq. (4) is the exact translation to canonical paper notation.

Quantity	Definition	Interpretation
Charge-density offset	$(\text{tr } G - V/2)/V$	Fraction of the total charge capacity by which a trajectory or Born mean differs from half filling.
Trajectory charge variance density	$\text{Var}_s(\text{tr } G)/V$	Extensive between-trajectory variance per single-particle mode.
Fractional-filling standard deviation	$\sqrt{\text{Var}_s(\text{tr } G)}/V$	Typical trajectory-to-trajectory deviation of the filling fraction.
Quantum charge variance density	$\text{tr}(G - G^2)/V$	Within-trajectory number uncertainty per mode.
RMS covariance norm	$\ Q\ _F / \sqrt{V}$	Intensive purity diagnostic with pure-state ceiling one.
Adjacent covariance step	$\ Q_t - Q_{t-1}\ _F / \sqrt{V}$	Trajectorywise motion per cycle; not a stationarity test by itself.

IV. GLOBAL CHARGE SHARPENING AND COVARIANCE PURIFICATION

A. Charge variance and its absolute cycle derivative

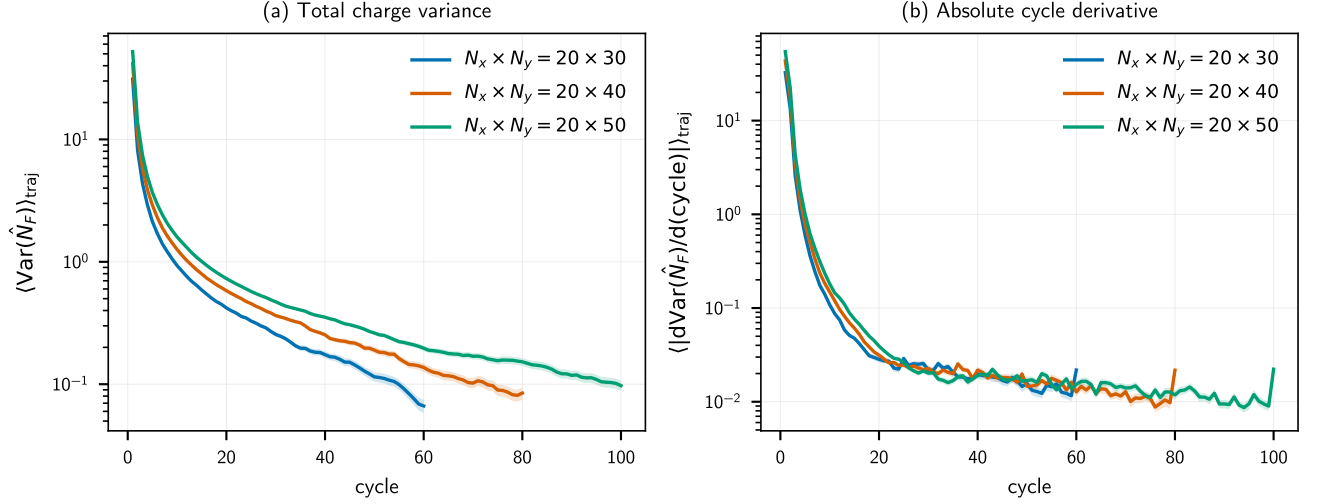


FIG. 1. Maximally mixed purification ensemble with 100 Born-rule trajectories. Left: sample-averaged within-trajectory quantum variance $\text{Var}(\hat{N}_F)$. Right: trajectory mean of the absolute cycle derivative in Eq. (8). Both y axes are logarithmic; the narrow bands are one standard error. No smoothing is applied. The domain wall and hard truncation are present in all three sizes.

Figure 1 shows a fast initial collapse followed by a much slower tail. At the final saved cycle, the unnormalized means are 0.06647, 0.08464, and 0.09733 for $N_y = 30, 40, 50$. The corresponding densities are 5.54×10^{-5} , 5.29×10^{-5} , and 4.87×10^{-5} per single-particle mode. The curves therefore grow weakly with N_y only because the observable is extensive; the final density slightly decreases over this range.

The absolute derivative falls from $O(10^1)$ in the first cycle to an $O(10^{-2})$ noisy late-time floor. Taking the absolute value before averaging is essential: a derivative

of the ensemble mean could hide trajectorywise motion by cancellation. The remaining floor also explains why convergence must be defined with a documented tolerance rather than the impossible condition of a numerically zero derivative.

B. Frobenius norm and operational global transient

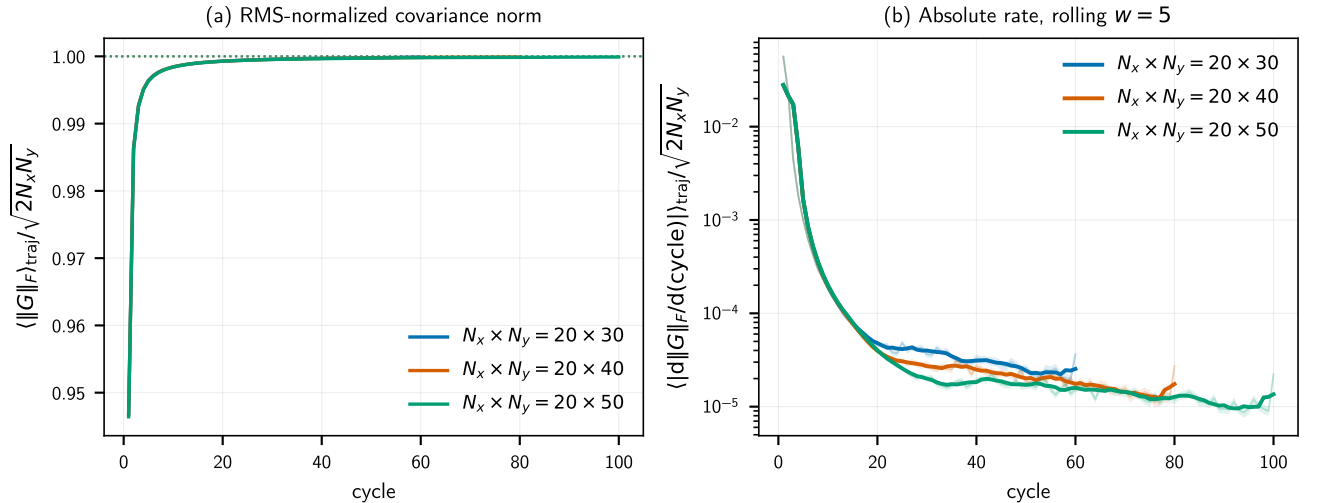


FIG. 2. The same maximally mixed ensemble expressed through the exact norm identity (14). Left: RMS-normalized covariance norm, whose pure-state ceiling is one. Right: its RMS-normalized absolute rate on a logarithmic axis. Faint curves and bands are unsmoothed means and standard errors; thick curves are centered rolling means with $W = 5$.

The apparent size dependence that resulted from dividing $\|Q\|_F$ by $2N_x N_y$ is absent in Fig. 2, because the correct

denominator is $\sqrt{2N_x N_y}$. All three curves approach the common pure-state ceiling. The late 20%-of-run mean

deficits are 1.61×10^{-4} , 1.23×10^{-4} , and 1.21×10^{-4} .

The operational global time is

$$T_{\text{conv}}^{(F)} = \max \left\{ T[d_F \leq 10^{-3}], T[r_F^{(W=5)} \leq 10^{-4}/\text{cycle}] \right\}, \quad (31)$$

where each inequality must remain satisfied at every later saved cycle. Trajectory bootstrapping with 2000 repetitions gives Table III.

TABLE III. Global covariance-purification convergence. Confidence intervals are percentile intervals from resampling complete trajectories.

N_y	$T[d_F \leq 10^{-3}]$	$T[r_F \leq 10^{-4}]$	$T_{\text{conv}}^{(F)}$	late $\bar{d}_F$	final $\text{Var}(\hat{N}_F)$
30	15 [15, 15]	14 [14, 14]	15 [15, 15]	1.61×10^{-4}	0.06647
40	15 [15, 16]	14 [14, 14]	15 [15, 16]	1.23×10^{-4}	0.08464
50	16 [15, 16]	14 [14, 14]	16 [15, 16]	1.21×10^{-4}	0.09733

Within $30 \leq N_y \leq 50$ and fixed $N_x = 20$, the global scalar transient is indistinguishable from an $O(1)$ cycle count with respect to N_y . This is a finite-range statement, not an asymptotic proof. More importantly, it says nothing by itself about a spatially localized slow mode: a boundary occupies a vanishing fraction of the full two-dimensional volume and can be hidden in a global

Frobenius average.

V. BORN-RULE TOTAL CHARGE: MEAN, SPREAD, AND STATIONARITY

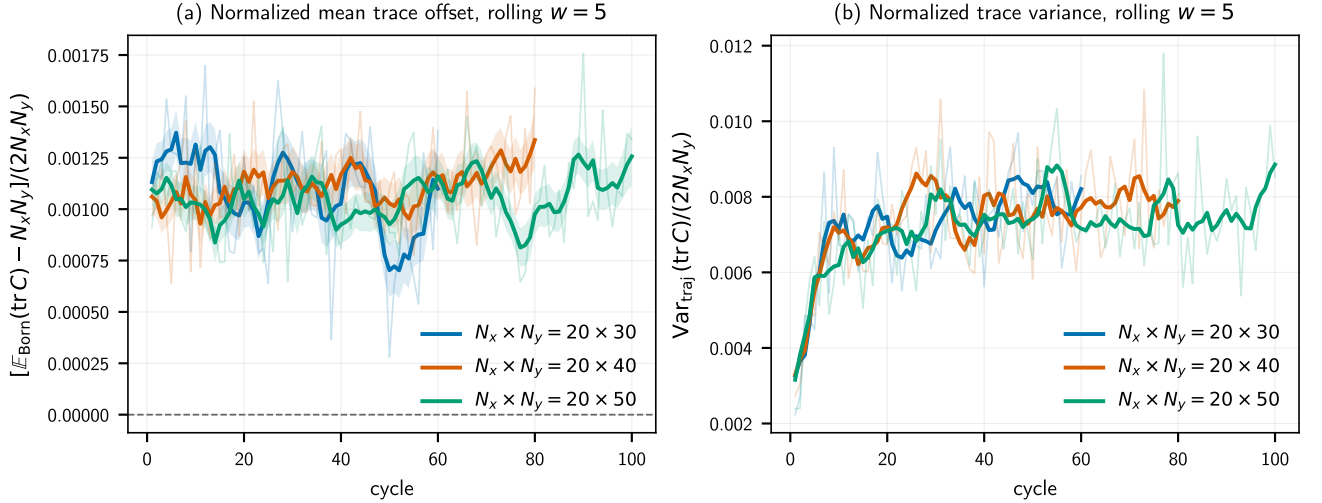


FIG. 3. Born-rule total-charge trace in the maximally mixed purification ensemble. Left: $[\overline{\text{tr } G} - V/2]/V$. Right: $\text{Var}_s(\text{tr } G)/V$. Thick lines are centered rolling means with $W = 5$; faint lines and bands show unsmoothed values and their standard errors. The nonzero right panel demonstrates that **perfect.correction** does not place all trajectories in one fixed-charge sector.

The ensemble mean remains close to half filling but not exactly at it. In the late half $N_y < t \leq 2N_y$, the mean fractional offsets are approximately 1.01, 1.14, and 1.09 parts in 10^3 . The trajectory variance density is nearly size

independent, about $7.6\text{--}7.9 \times 10^{-3}$. The typical fractional-filling deviation in Eq. (12), however, decreases as $V^{-1/2}$ from 2.56×10^{-3} to 1.95×10^{-3} .

TABLE IV. Late-window total-charge and quantum-variance statistics for the maximally mixed ensemble, averaged over cycles $N_y + 1$ through $2N_y$. σ_n/V is the typical deviation of the fractional occupancy across trajectories.

N_y	$\bar{q}$	$\text{Var}_s(n)/V$	σ_n/V	$\overline{\text{Var}(\hat{N}_F)}$	$\overline{\text{Var}(\hat{N}_F)}/V$
30	1.007×10^{-3}	7.872×10^{-3}	2.561×10^{-3}	0.1464	1.220×10^{-4}
40	1.139×10^{-3}	7.830×10^{-3}	2.212×10^{-3}	0.1455	9.092×10^{-5}
50	1.087×10^{-3}	7.634×10^{-3}	1.954×10^{-3}	0.1607	8.034×10^{-5}

The expected occupancy is therefore tightly bounded empirically around half filling: the mean bias is about 0.1% of capacity and a typical trajectory is within roughly 0.2–0.26% of capacity. These are finite-sample statistics, not rigorous probability bounds. They can support a statement about the observed ensemble but cannot prove confinement for all times or sizes.

The slowly falling quantum variance in Fig. 1 supplies a second caution. In the maxmix ensemble, its first-to-second half changes over $[N_y, 2N_y]$ are -7.31×10^{-5} , -5.49×10^{-5} , and -3.47×10^{-5} after division by V , with bootstrap intervals that exclude zero. Thus the expected total charge is approximately stationary by N_y , but the state is still purifying weakly. The entire window must not be called a stationary charge-variance ensemble without this qualification.

VI. WHERE THE CHARGE FLUCTUATIONS LIVE

A. Three inequivalent spatial profiles

Define the cell occupation

$$n_s(x, y, t) = \sum_{\mu=1}^2 [G_s(t)]_{(x,y,\mu),(x,y,\mu)}. \quad (32)$$

At half filling its reference value is one. The late-time spatial analysis uses cycles $N_y + 1$ through $2N_y$, averages along y , and keeps separate:

$$\begin{aligned} b(x) &= \left| \overline{n(x, y, t)} - 1 \right|, \\ v_{\text{traj}}(x) &= \overline{[n_s(x, y, t) - \overline{n(x, y, t)}]^2}, \\ v_q(x) &= \overline{\text{Var}_s[\hat{N}_{x,y}]}. \end{aligned} \quad (33)$$

The first is ensemble bias, the second is between-trajectory fluctuation of an expectation value, and the third is quantum fluctuation inside each trajectory.

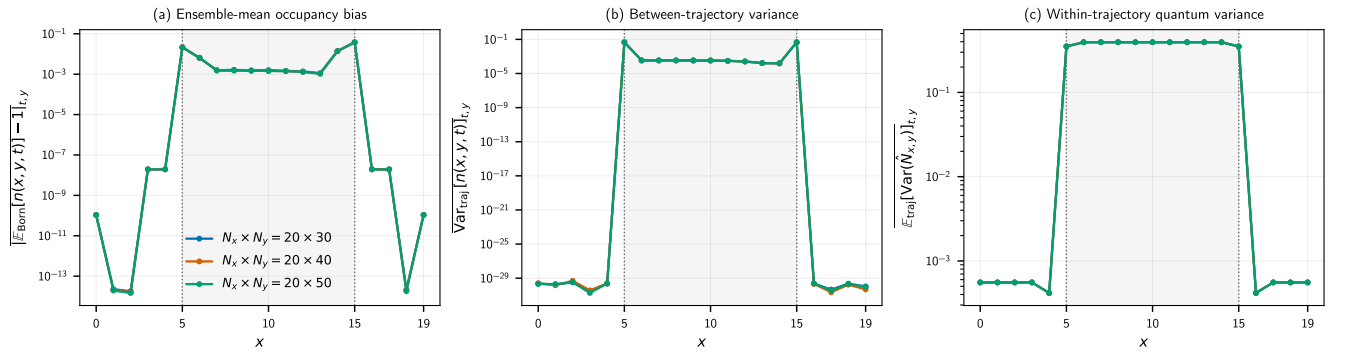


FIG. 4. Late-window charge profiles in the $N_x = 20$ maxmix ensemble. Panel (a) shows the absolute bias of the Born mean from one particle per cell; panel (b) shows between-trajectory variance of the expected cell occupancy; panel (c) shows the within-trajectory quantum variance. The shaded active slab is $5 \leq x \leq 15$, and the dotted lines are its two domain walls. Curves for all three N_y values nearly collapse.

Figure 4 supports a strong but carefully delimited wall statement. The ensemble-mean bias and especially the

trajectory-to-trajectory variance are wall localized. The within-trajectory quantum variance is not: it is broad and nearly uniform across the active slab.

TABLE V. Late-window wall localization. “Exact wall” uses $x = 5, 15$; “distance ≤ 1 ” adds their neighboring rows. The wall/bulk density ratio uses wall-adjacent rows $x = 5, 6, 14, 15$ and active-bulk rows $x = 7, \dots, 13$.

N_y	Metric	exact-wall capture	distance- ≤ 1 capture	wall/bulk ratio
30	mean occupancy bias	0.657	0.884	13.36
40	mean occupancy bias	0.668	0.891	14.30
50	mean occupancy bias	0.665	0.892	14.40
30	trajectory occupancy variance	0.973	0.978	77.91
40	trajectory occupancy variance	0.972	0.978	76.88
50	trajectory occupancy variance	0.973	0.978	78.00
30	within-trajectory quantum variance	0.165	0.351	0.946
40	within-trajectory quantum variance	0.165	0.350	0.945
50	within-trajectory quantum variance	0.165	0.350	0.945

More than 99.9999% of the first two profiles lies inside the active slab. That near-zero exterior is partly imposed by hard domain-wall truncation and must not be presented as an emergent proof of localization. The robust interior statement is instead that approximately 97.3% of the between-trajectory occupancy variance lies on the two exact wall rows and that the wall-row variance density is about 77–78 times the active-bulk value. The phrase “most charge fluctuations come from the boundary”

is therefore correct for ensemble-to-ensemble occupancy fluctuations, but false for the within-trajectory quantum charge variance.

B. Reduced-covariance pilot

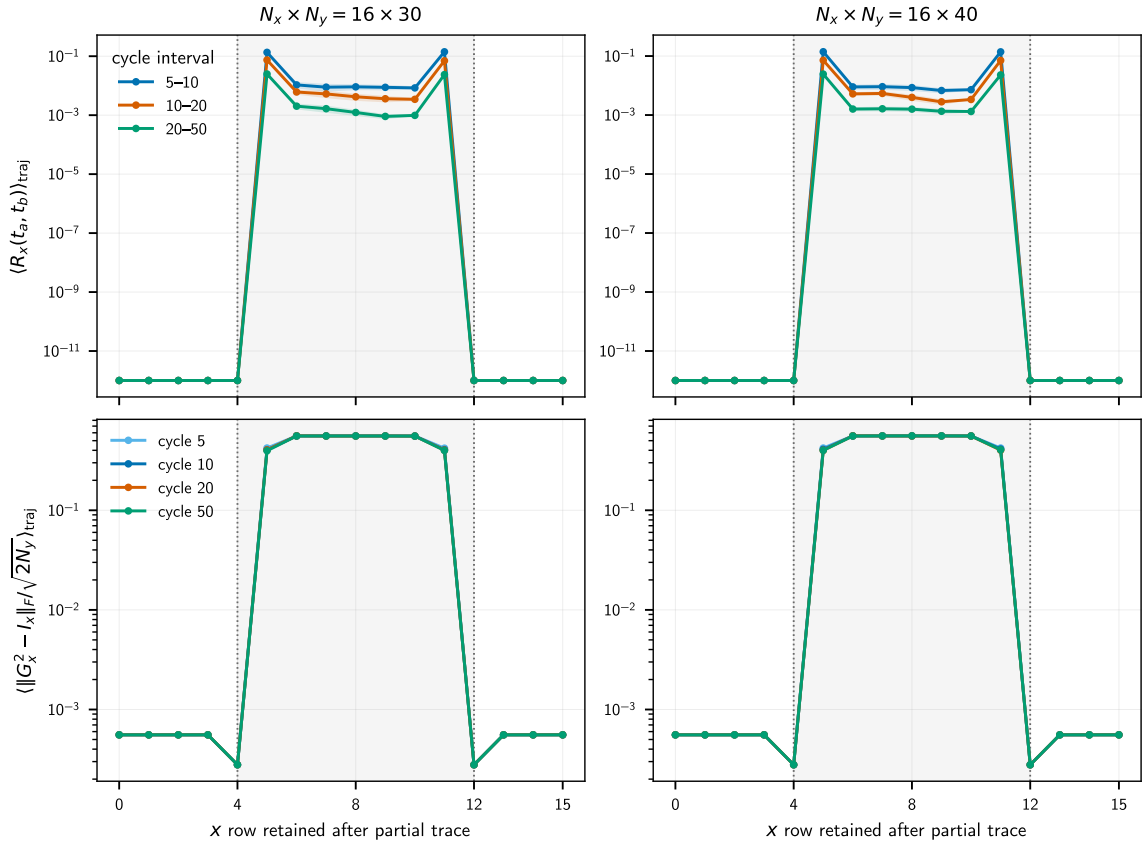


FIG. 5. Sparse-snapshot pure-state pilot for $N_x = 16$. Top row: the reduced covariance rate $R_x(t_a, t_b)$ of Eq. (21); bottom row: reduced-state mixedness $M_x(t)$ of Eq. (22). Dotted lines mark the walls at $x = 4, 12$. The rate is 7.5–8.9 times larger on active wall rows than in the active bulk over the three observed intervals. Ten trajectories and only four snapshot cycles make this descriptive rather than a precision scaling study.

The top panels of Fig. 5 give the desired spatial resolution of covariance motion after tracing out all other x

rows. In the 20–50 interval the wall-to-bulk rate ratios are 8.94 for 16×30 and 7.92 for 16×40 . The active-bulk rate has already fallen to 1.36×10^{-3} and 1.49×10^{-3} per cycle, whereas the wall remains at 1.22×10^{-2} and 1.18×10^{-2} .

The bottom panels explain why reduced-state mixedness cannot replace a temporal rate. M_x remains $O(1)$ inside the active slab even while successive curves nearly coincide, because tracing out the other rows creates a mixed subsystem. Mixedness is a property of the reduced

state; R_x is the finite-difference convergence diagnostic.

VII. FAST BULK AND ALGEBRAIC WALL PURIFICATION

A. Sparse covariance-rate evidence

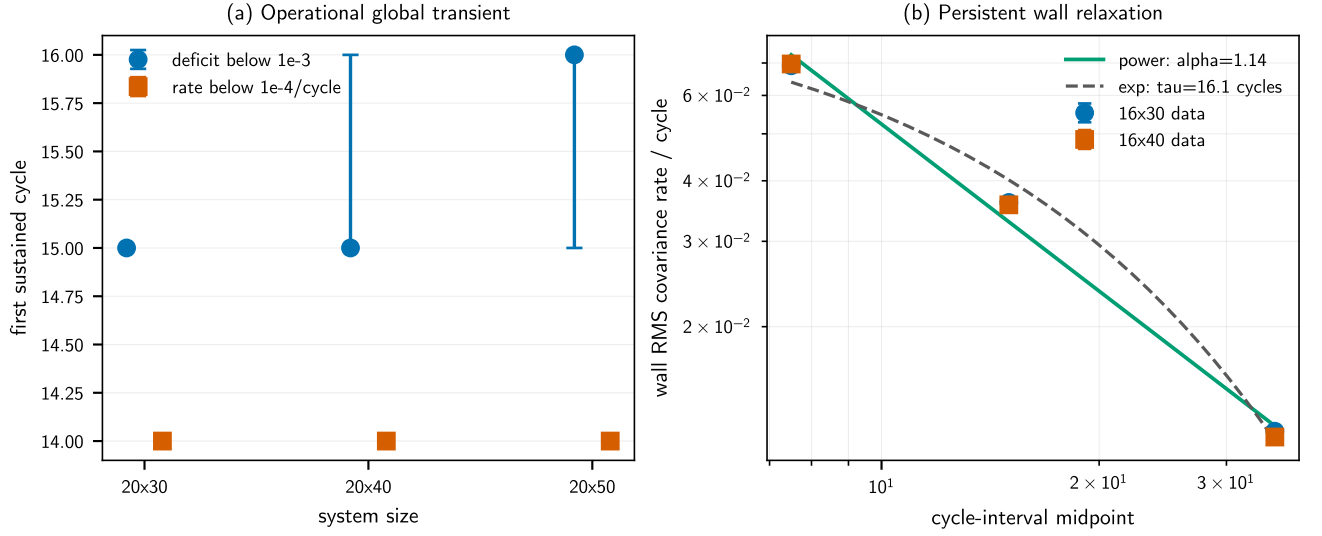


FIG. 6. Two complementary convergence summaries. Left: the global Frobenius thresholds defining Eq. (31). Right: common-amplitude fits of the six sparse wall-rate points from the reduced-covariance pilot to a power law and an exponential. The wall-rate power fit gives $\alpha = 1.144$; the exponential gives $\tau = 16.1$ cycles.

The sparse R_x pilot favors a power law over an exponential by $\Delta\text{AICc} = 3.52$; in 2000 paired trajectory bootstraps, the power law has lower AICc in 97.45% of resamples. The bootstrap interval for the model difference narrowly includes zero, so this is supporting rather than decisive evidence. The full entropy-contour histories

provide the stronger test.

B. Cycle-resolved entropy contours

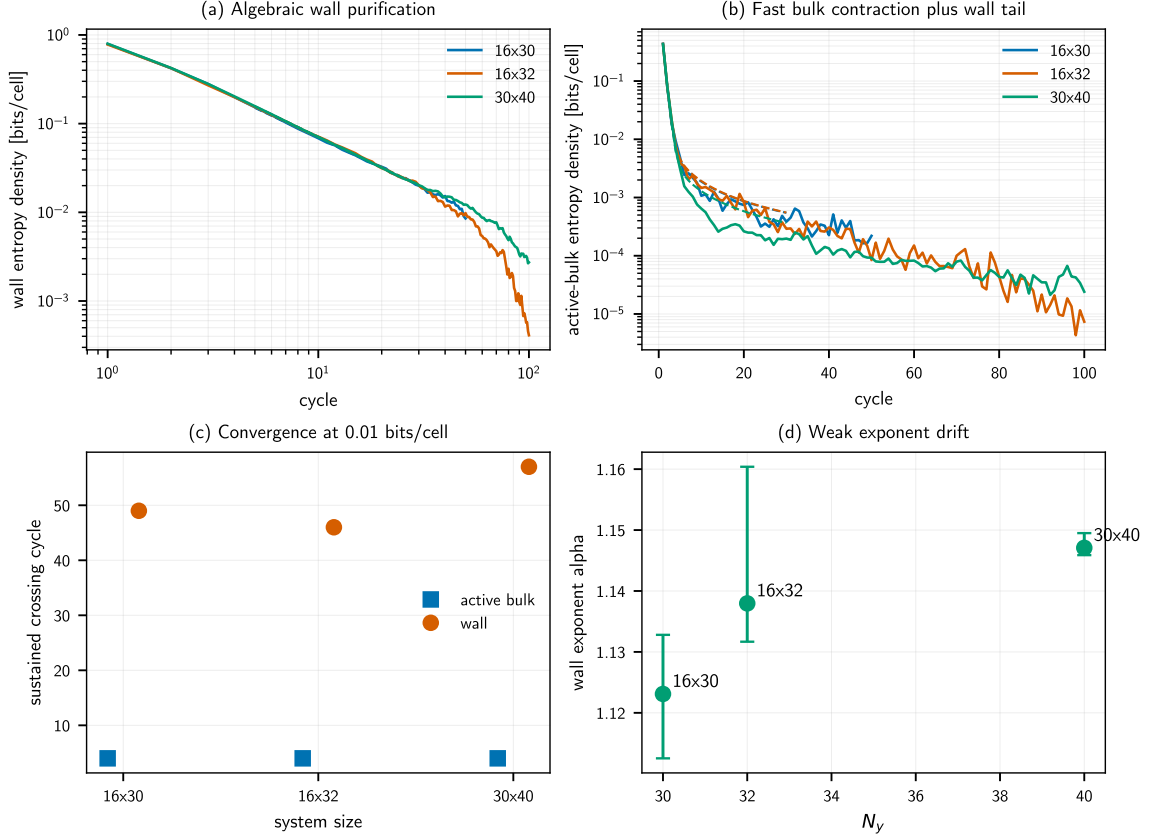


FIG. 7. Spatially resolved purification from maximally mixed initial states. (a) Wall entropy density on log-log axes. (b) Active-bulk entropy density on a logarithmic y axis. (c) first sustained crossing of 0.01 bits per cell. (d) wall power-law exponent from cycles 5–30; error bars show the range obtained by moving the fit onset among cycles 3, 5, 8, 10, not a statistical confidence interval. The 16×30 run ends at cycle 50; the other two end at cycle 100.

The wall curves in Fig. 7(a) are nearly straight over more than a decade on log-log axes. Direct wall fits give $\alpha_w = 1.1231, 1.1380, 1.1471$, with log-space $R^2 = 0.99929, 0.99847, 0.99989$ for the three geometries. Replacing the power law by an exponential raises AICc by

112.9, 91.1, and 138.8, respectively. In contrast, the intrinsic bulk component in Eq. (25) has $\tau_b = 0.564, 0.570, 0.585$ cycles. The subsequent weak bulk tail is well described as leakage of the same wall power law into the regional average.

TABLE VI. Wall/bulk purification statistics. The $2N_y$ entropy density is reported at cycle 50 for 16×30 because that run was not saved to cycle 60. A dash denotes a right-censored threshold.

Geometry	α_w	$\Delta\text{AICc}_{\text{exp-pow}}$	τ_b	$T_w(10^{-2})$	$T_b(10^{-2})$	s_w near $2N_y$
16×30	1.123	112.9	0.564	49	4	8.52×10^{-3} at $t = 50$
16×32	1.138	91.1	0.570	46	4	4.57×10^{-3} at $t = 64$
30×40	1.147	138.8	0.585	57	4	4.85×10^{-3} at $t = 80$

These data explain why there is no unique “the transient time.” A global norm gives 15–16 cycles; active-bulk entropy gives four cycles at the 0.01-bit tolerance; the wall requires about 50–60 cycles. Because the wall decay is algebraic, its tolerance-dependent time behaves approximately as

$$T_w(\epsilon) \sim \left(\frac{A_w}{\epsilon} \right)^{1/\alpha_w}, \quad (34)$$

rather than being characterized by one exponential time

constant.

The existing geometries do not determine an asymptotic system-size exponent for T_w . Only 16×30 and 16×32 hold N_x fixed, and their N_y range is too narrow. The 30×40 case changes both dimensions. Late finite-size crossover fits in the longer runs give descriptive scales 20.35 and 34.74 cycles and crossover cycles 53 and 82, but those two points cannot establish a law. What is supported is a robust separation between rapid bulk contraction and persistent algebraic wall relaxation, consistent with the independent purification plots.

C. Is cycle $2N_y$ late enough?

For the operational tolerance $s_w \leq 0.01$ bits per cell, yes for every available entropy-contour geometry. The crossings occur by cycle 57, while the relevant endpoints are 60, 64, 80; the 16×30 file stops at cycle 50 but has already crossed at cycle 49. At the more stringent 10^{-3} threshold, 16×32 crosses only at cycle 93, while the other

two wall curves are right-censored. The statement “ $2N_y$ is sufficient” must therefore include the observable and tolerance. It is supported at the 0.01-bit level, not at arbitrary precision.

VIII. A TRAJECTORY DOES NOT STOP MOVING AT CONVERGENCE

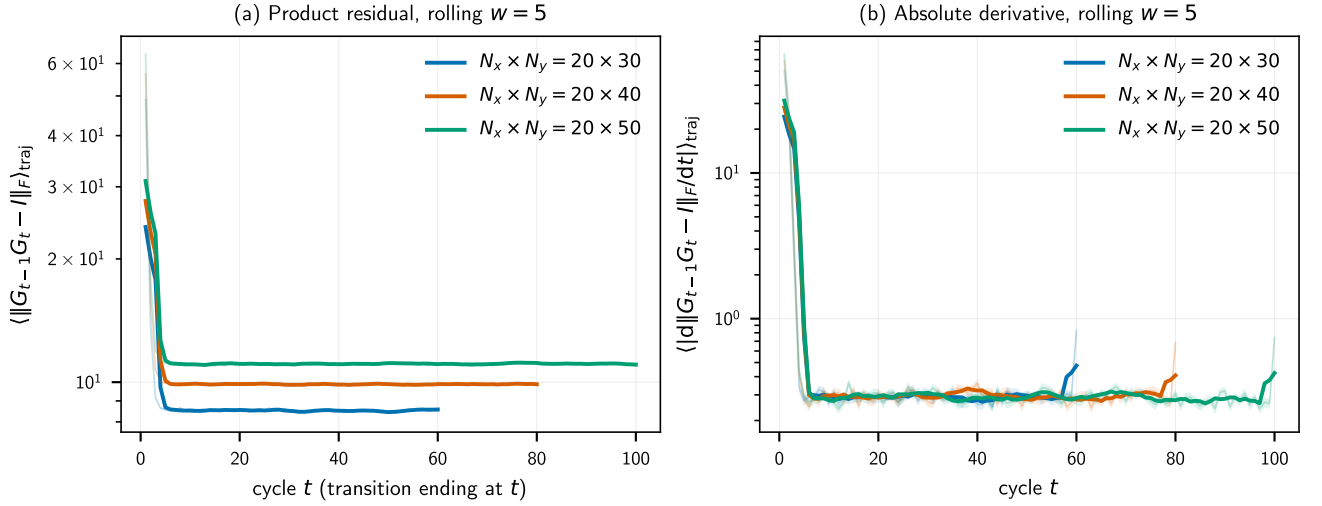


FIG. 8. Pure-state default-initialized streaming ensemble. Left: $\overline{\|Q_{t-1}Q_t - \mathbf{1}\|_F}$, exactly equal to $\overline{\|Q_t - Q_{t-1}\|_F}$ by Eq. (19). Right: the trajectory mean absolute cycle derivative. Thick curves use a centered rolling mean with $W = 5$; faint curves and bands are unsmoothed. Both axes are logarithmic.

After the initial drop, the adjacent-cycle residual reaches a size-extensive plateau: its late means are 8.52, 9.89, and 11.09. Dividing by $\sqrt{V}$ gives 0.24597, 0.24730, and 0.24787. This near collapse shows $O(1)$ RMS covariance motion per cycle even at late times. Its derivative becomes small because the *magnitude* of the motion stabilizes; the state itself does not stop changing.

This observation is fully compatible with convergence to a steady ensemble. A Markov chain at stationarity keeps moving between states while preserving its probability

distribution. Therefore $\|Q_t - Q_{t-1}\|_F \rightarrow 0$ is neither required nor expected. Convergence should instead be tested by stationarity of selected distributions and by their temporal autocorrelation.

IX. BURN-IN, STATIONARITY, AND EFFECTIVE ENSEMBLE SIZE

A. Cycle-resolved post-burn-in behavior

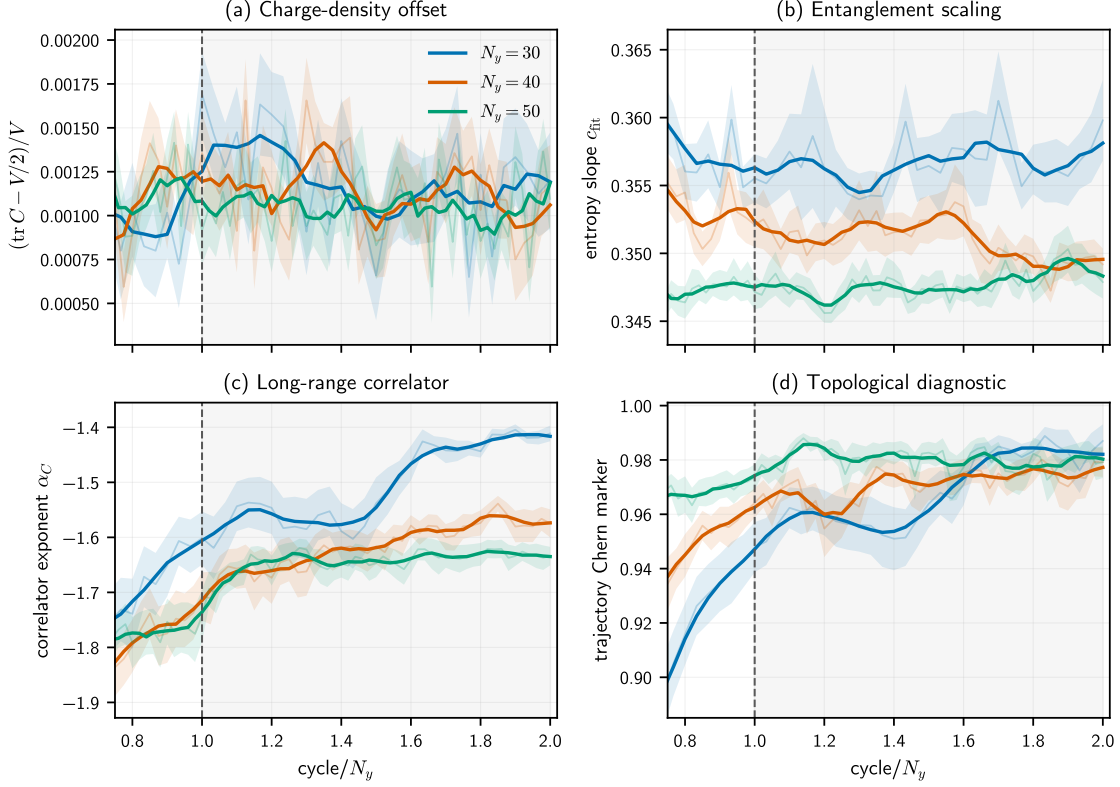


FIG. 9. Default pure-state ensemble near and after the proposed burn-in. Cycle number is scaled by N_y ; the dashed line marks $t = N_y$ and the shaded region is the intended measurement window. Faint curves and bands show raw trajectory means and standard errors; thick curves are centered rolling means with $W = 5$ for display only. The formal block tests use unsmoothed data. The charge density and entanglement slope are stable, while the smallest sizes retain visible long-range-correlator and Chern-marker drift.

The four panels in Fig. 9 already show why no single scalar certifies the entire covariance distribution. The charge density is stable to the 10^{-4} – 10^{-3} scale and the entropy slope is visually flat. The correlator exponent continues to relax upward, most clearly for $N_y = 30$, and the Chern marker still rises slightly in the same system.

The paired block test makes this quantitative. Table VII reports raw first-to-second-half differences and

the integrated autocorrelation time. The $N_y = 30$ charge-density interval narrowly excludes zero; its effect is small, a shift of 2.11×10^{-4} of total capacity. Entropy-slope intervals include zero for every size. Correlator-exponent intervals exclude zero for $N_y = 30, 40$ but include zero for $N_y = 50$. The Chern-marker change is detected only for $N_y = 30$. The normalized covariance-step magnitude is stationary for all three sizes.

TABLE VII. Post-burn-in stationarity and correlation in the default pure-state ensemble. Differences are second-half minus first-half means within $[N_y, 2N_y]$; brackets are trajectory-bootstrap 95% intervals. The last two columns give integrated autocorrelation times in cycles.

N_y	Δc_{fit}	$\Delta \alpha_C$	ΔC_{RS}	$\tau_{\text{int}}[c_{\text{fit}}]$	$\tau_{\text{int}}[\alpha_C]$
30	+0.00114 [−0.00250, 0.00502]	+0.1289 [0.0672, 0.1984]	+0.02327 [0.01106, 0.03696]	5.04	14.81
40	−0.00118 [−0.00342, 0.00105]	+0.0689 [0.0127, 0.1293]	+0.00713 [−0.00256, 0.01717]	5.60	20.25
50	+0.000915 [−0.000329, 0.002322]	+0.0207 [−0.0089, 0.0498]	−0.00162 [−0.00793, 0.00450]	6.60	24.10

B. Stationary does not mean independent

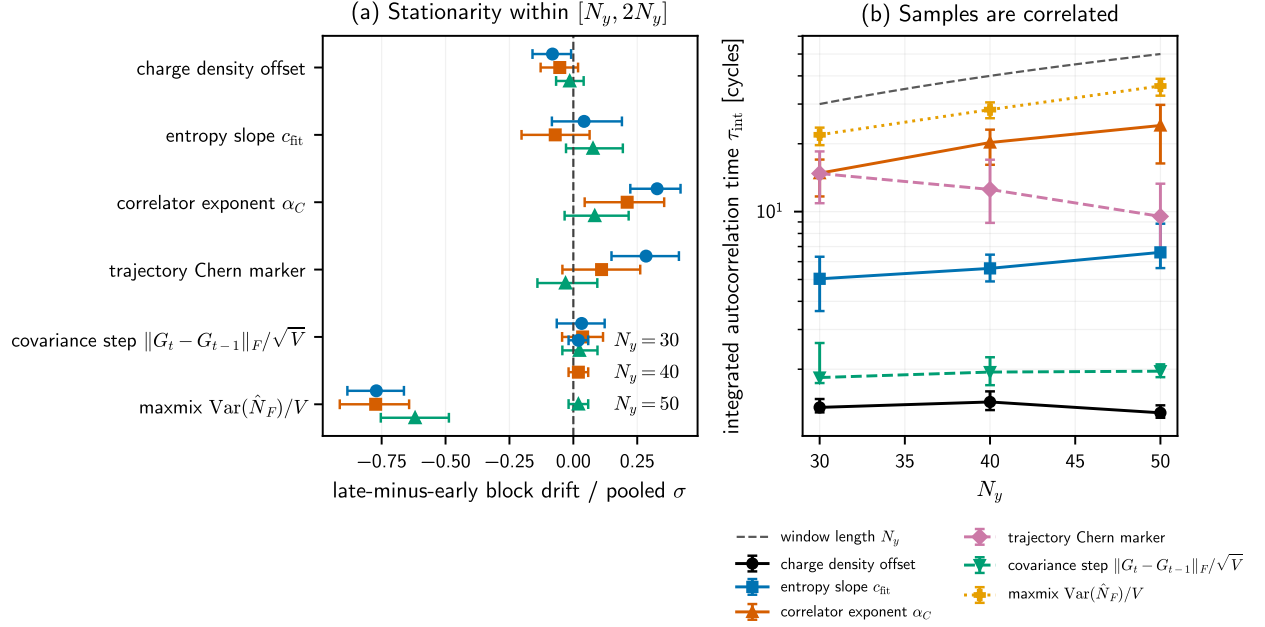


FIG. 10. Quantitative ensemble test. Left: second-minus-first-half block drift divided by the pooled within-window standard deviation; horizontal bars are trajectory-bootstrap 95% intervals. The maxmix quantum charge variance is included to show that purification remains nonstationary. Right: integrated autocorrelation times from trajectory residuals after removing the ensemble mean separately at each cycle. The dashed line is the N_y -cycle measurement window length.

The autocorrelation hierarchy in Fig. 10(b) is physically informative. The pure-state total-charge density has $\tau_{\text{int}} = 1.35, 1.43, 1.28$ cycles; the covariance-step magnitude has 1.84, 1.94, 1.96 cycles. The entanglement slope is correlated over 5.04, 5.60, 6.60 cycles. The long-range correlator exponent is much slower:

$$\tau_{\text{int}}[\alpha_C] = (14.81, 20.25, 24.10) \simeq (0.494, 0.506, 0.482)N_y. \quad (35)$$

The three points are consistent with a roughly linear dependence over the available range, although they are not enough for a controlled dynamical exponent. This slow temporal mixing is the ensemble counterpart of the algebraic wall purification.

Over the inclusive windows of 31, 41, 51 cycles, the effective counts per trajectory are therefore about 6.16, 7.32, 7.73 for the entropy slope but only 2.09, 2.02, 2.12 for the correlator exponent. Multiplying by 100 independent trajectories gives roughly 616, 732, 773 and 209, 202, 212 independent-equivalent samples, respectively. Every retained cycle improves an estimator, but adding one cycle does not add one independent wall sample.

For the maxmix quantum charge variance, $\tau_{\text{int}} = 21.9, 28.3, 36.1$ cycles and the standardized early-to-late drift is large. This is precisely the behavior of an ongoing purification transient. Its cycles should not be pooled as a steady ensemble even though the same cycle range is convenient for displaying purification.

C. What can be claimed about the ensemble

The evidence supports the following statement:

After a burn-in, cycle-resolved states are retained as samples of a temporally correlated Born-rule ensemble. Stationarity is checked by comparing early and late blocks of the measurement window, and statistical uncertainties are computed by resampling trajectories or otherwise accounting for integrated autocorrelation. Entanglement observables are stationary after an N_y burn-in for the tested sizes, while the smallest systems retain measurable drift in long-wavelength wall correlations.

This claim is empirical and observable-resolved. It does not establish equality of the full probability distribution over all covariance matrices, which would require a much more demanding two-sample test in a high-dimensional state space.

X. A SYSTEMATIC CONVERGENCE PROTOCOL

A good convergence timescale should be defined by the observable and the target precision, not selected from a visually flat curve. The following procedure is supported by the present data.

1. Choose observables that cover the sectors relevant to the final claim: a global scalar, an active-bulk

diagnostic, a wall-localized diagnostic, and any topological or long-range correlation statistic that will appear in the paper.

2. For each observable O , choose a physically interpretable tolerance ϵ_O . Define T_O by a sustained threshold or, for a quantity with a nonzero plateau, by a sustained rate threshold.
3. Set the provisional burn-in to

$$T_{\text{burn}} = \max_O T_O. \quad (36)$$

For an algebraic wall mode, document the tolerance explicitly because Eq. (34) makes T_O tolerance dependent.

4. On the proposed measurement window, compare its first and second halves using paired trajectory block means. Increase the burn-in if a practically important observable exhibits statistically resolved drift.
5. Estimate $\tau_{\text{int}}[O]$ and report either trajectory-block bootstrap errors or N_{eff} . Select the measurement-window length from the slowest relevant τ_{int} , not from the number of saved cycles alone.
6. Repeat at multiple N_y while holding the other geometry fixed before claiming a scaling law for T_{burn} or τ_{int} .

Applied here, this protocol distinguishes three uses of N_y :

- As a *burn-in*, N_y is conservative for $N_y = 50$ and for global/entanglement observables. For $N_y = 30, 40$, it begins before the slowest wall sector is fully stationary according to the correlator exponent. A conservative finite-size rule for the present dataset is $T_{\text{burn}} \gtrsim \max(N_y, 50-60)$ when long-wavelength wall observables matter.
- As a *measurement-window length*, N_y is well motivated because the slow correlator has $\tau_{\text{int}} \approx 0.49N_y$, yielding about two independent-equivalent wall samples per trajectory. A shorter $O(1)$ window would lose effective wall statistics as N_y grows.
- As an *endpoint*, $2N_y$ is sufficient for the observed 0.01-bit wall-entropy criterion and clearly sufficient for the global Frobenius criterion. It is not an assertion of arbitrary-precision convergence.

This also answers whether cycle time must scale with N_y . The global purification burn-in need not scale with N_y over the tested range. The wall's temporal autocorrelation, however, is approximately proportional to N_y , so the amount of simulation required for a fixed number of independent-equivalent wall samples does scale with N_y . Burn-in and sampling cost are different questions.

XI. INTERPRETATION AND LIMITATIONS

The combined evidence supports a physical picture of fast bulk contraction and a slowly purifying critical wall. Global norms rapidly approach their pure ceiling because the bulk dominates the volume. Boundary-resolved entropy and reduced-covariance motion retain algebraic tails. Once the marginal distribution has settled, the circuit keeps generating distinct trajectory states; long-wavelength wall information remains correlated across many cycles.

Several limitations constrain the language:

1. The $N_x = 20$ global campaign has only three circumferences. Constancy of $T_{\text{conv}}^{(F)}$ over $N_y = 30-50$ is not an asymptotic theorem.
2. The spatial covariance pilot has ten trajectories and four snapshots. Its R_x ratios are strong descriptive evidence, not a fitted dynamical exponent.
3. The wall-entropy geometries do not form a clean one-parameter size series. They establish wall/bulk separation and power-law time dependence, but not a system-size scaling exponent for the wall threshold time.
4. Hard domain-wall truncation enforces much of the near-zero exterior. Claims should concern localization within the active slab unless compared against an untruncated control.
5. Stationarity tests cover selected observables, not the full covariance distribution. Failure to detect drift is not proof of exact stationarity.
6. The centered rolling average with $W = 5$ is a visualization aid. It is not additional data, and formal block comparisons use raw cycle values.

The most important semantic limitation is the distinction between nonlinear trajectory observables and observables of the averaged correlation matrix. For example, $\overline{C_G}$, the Born average of trajectory-resolved Chern numbers, is generally not $C_{\overline{G}}$, the Chern number computed after averaging G . The present Chern stationarity test uses the former.

XII. REPRODUCTION CHECKLIST

The authoritative analysis root is `analysis_outputs/total_charge_variance_cycle_derivative/` within the `colab_charge_fluctuations` package. Its manifest records the complete source-file list, estimator definitions, campaign identifiers, and output paths. The original analysis notebook is `plot_total_charge_variance_and_derivative_cpu.ipynb`.

The two stationarity figures and their tables are regenerated locally by `make_ensemble_sampling_figures.py`.

The script reads the archived streaming fits, scalar metrics, local charge arrays, and maximally mixed total charge variance; it does not rerun dynamics. It checks a complete $100 \times 2N_y$ grid for every observable and size, uses 2000 trajectory bootstrap repetitions with seed 314159, and writes:

- `post_burnin_stationarity_time_series.csv`,
- `post_burnin_stationarity_autocorrelation_summary.csv`,
- the corresponding PDF and PNG figure pairs, and
- `analysis_manifest.json` beside this note.

All other figures are the archived PDFs emitted by the characterization notebook. Compilation uses RevTeX 4.2 in two-column PRB format.

XIII. CONCLUSIONS

The charge and covariance data admit a coherent but nontrivial convergence story. Global purification is fast and essentially independent of N_y over the explored range. The active bulk contracts exponentially on an $O(1)$ cycle scale. The domain wall purifies algebraically and controls the longest transient and autocorrelation scales. Ensemble-mean occupancy bias and trajectory-to-trajectory occupancy fluctuations are sharply wall localized, while within-trajectory quantum charge variance is spread throughout the active slab.

The production convention of burning in for N_y cycles and measuring on $[N_y, 2N_y]$ is therefore sensible, but its meaning should be stated precisely. At cycle $2N_y$, the observed wall entropy is below 0.01 bits per cell and global covariance purification is long complete. During the measurement window, each cycle can be retained as a Born-rule ensemble sample once the observable's marginal distribution passes a stationarity check. The cycles are not independent: the slow correlator contributes only about two independent-equivalent samples per trajectory over an N_y -cycle window. This distinction permits use of all saved cycles without overstating the statistical sample size.

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[2] P. Calabrese and J. Cardy, Entanglement entropy and quantum field theory, *J. Stat. Mech.* P06002 (2004).
 [3] A. D. Sokal, Monte Carlo methods in statistical mechanics: Foundations and new algorithms, in *Functional Integration: Basics and Applications* (Plenum, New York, 1997).